Zhou Yang, PhD

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EXPERIENCE

Independent Researcher, Health Economics and Health AI
Jan 2026 to now

Toronto

- Working papers on how AI in health care gets tested, how it gets paid for, and how patients use it. See early findings.

Data Scientist, Certilytics
Sep 2022 to Sep 2026

Remote

- Led 40+ causal evaluations across 15+ healthcare programs, using claims, eligibility, utilization and cost data.
- Presented findings and recommendations to client executives and state Medicaid agency staff.
- Built cost-benefit, ROI and budget-impact models that informed renewal and program decisions.
- Built cost-effectiveness models (Markov state-transition models, QALYs) and survival models (Weibull, Cox) for long-term cost and outcome estimates.
- Helped build a GenAI analytics assistant and the evaluation harnesses that tested AI outputs before release.
- Fine-tuned a biomedical language model (BioBERT) to turn unstructured clinical text into structured outcome variables.
- Deployed machine learning models in many projects, and built risk-prediction and forecasting models.
- Supervised three junior colleagues and led teams of three PhD researchers on public-health topics.

Data Scientist (Contract), Oracle
May to Dec 2021

Seattle

- Built deep-learning models that combine image and text features to predict product characteristics.
- Co-developed a multimodal double machine learning framework for apparel price elasticity.

Economist (Contract), Amazon
May to Dec 2020

Seattle

- Estimated cross-product elasticity across 1.2M+ products with econometric and machine learning methods.
- Built a product valuation model to prioritize supply-chain products by consumer demand and value.

Research Assistant (Part-time), University of Washington
Jan 2018 to Mar 2022

Seattle

- Estimated how Amazon Fresh's entry affected grocery stores (staggered difference-in-differences across 8,380 stores in 17 markets) and how fulfillment centres affected local retail jobs (instrumental variables).

Graduate Instructor, Economics, University of Washington
2019 to 2022

Seattle

- Taught 8 to 9 sections of introductory economics for about 300 students.

EDUCATION

PhD, Economics, University of Washington
2023

Dissertation: Essays on the Digital Transformation of Retail Grocery Industry. Chair: Yuya Takahashi.

BSc, Mathematics, Augsburg University
2017

WORKING PAPERS

Who checks the checkers? The price of not knowing reviewer accuracy in AI testing. Preprint, not peer reviewed. SSRN, November 2026: ssrn.com/abstract=7682921, DOI 10.2139/ssrn.7682921.

Access or reliance? Using AI chatbot advice instead of contacting a health professional among Canadian health-chatbot users. Preprint, not peer reviewed. SSRN, November 2026: ssrn.com/abstract=7683188, DOI 10.2139/ssrn.7683188.

Who gets the time AI saves? In preparation.

PUBLICATIONS AND PATENTS

Ren L, Yang Z, Wang Y, et al. The relations among worry, meta-worry, intolerance of uncertainty and attentional bias for threat in men at high risk for generalized anxiety disorder: a network analysis. *BMC Psychiatry*. 2020;20(1):452.

Co-inventor. Machine Learning Model and Neural Network to Predict Object Characteristics from Digital Image Similarities. US patent application US20230401589A1.

SKILLS

Causal inference	Difference-in-differences (including staggered), synthetic control, regression discontinuity, instrumental variables, propensity score matching, doubly robust estimation, double machine learning, causal forests
Health economics	Cost-effectiveness analysis (Markov state-transition models, QALYs, ICERs), budget impact models, survival analysis (Cox, Weibull)
Experiments	Online A/B tests on price design
Machine learning	Risk prediction, forecasting, gradient-boosted trees, deep learning in PyTorch
Language models	Retrieval-augmented generation, fine-tuning (LoRA, QLoRA), biomedical named entity recognition (BioBERT), evaluation harnesses, Hugging Face, LangChain
Health data	Claims, eligibility and utilization data; OMOP CDM; HL7 FHIR; risk adjustment
Tools	Python, R, SQL, SQL Server, Snowflake, PySpark, AWS (SageMaker, Redshift), Docker, Git, Power BI, Excel and VBA